

Exploratory Data Analysis: Customer and Segment Patterns

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```
# Load packages needed for the future EDA report.
knitr::opts_chunk$set(
  echo = FALSE,
  warning = FALSE,
  message = FALSE
)
```

Introduction

Dataset Overview

Customer Profile Analysis

Customer age is explored as a foundational customer-profile characteristic. This descriptive comparison examines whether age distributions differ between fraud and non-fraud transactions in the balanced analysis sample; it does not imply that customer age causes fraud.

Table 1: Customer-age summary by fraud status

fraud_status	sample_size	mean_age	median_age	standard_deviation	minimum_age	maximum_age
Non-fraud	1000	42.19	42	13.47	18	83
Fraud	1000	41.85	42	13.35	18	85

The following faceted histogram provides a visual comparison of customer-age distributions between the two fraud-status groups.

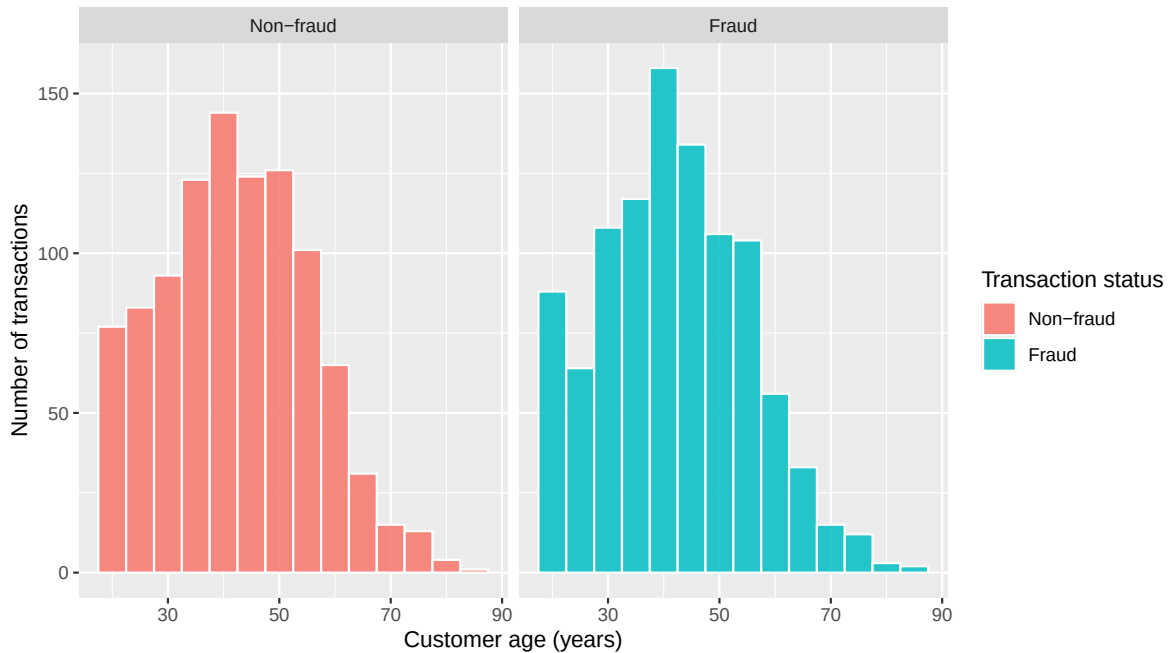


Figure 1: Distribution of customer age by fraud status in the balanced analysis sample.

The figure compares customer-age distributions between fraud and non-fraud groups in a sample balanced by design. It is intended only for descriptive comparison and should not be used to draw causal conclusions or estimate population fraud prevalence.

Customer Behavior Analysis

Geographic Analysis

Multivariate Segment Analysis

Key Findings

Business Insights

Conclusion